James Kent

Research Associate

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Education

2014 – 2020 **PhD. Neuroscience**, *University of Iowa*, Iowa City, IA
Research Focus: fMRI Methods/Reproducible Research
Advisor: Dr. Michelle W. Voss

2014 **B.A. Major: Psychology & Concentration: Neuroscience**, *Grinnell College*, Grinnell, IA

Appointments

2020 – Present **NeuroSynth 2.0 Technical Lead**, *University of Texas at Austin / Neuroinformagic Lab*, Austin, TX

Lead development and release engineering across NeuroStore and Neurosynth Compose. Built restartable, fingerprint-cached literature ingestion workflows linking article retrieval, structured table and prose extraction, metadata, and studyset delivery; maintain the OpenAPI specifications and Python/TypeScript SDK release pipeline. Recent releases include NeuroStore v0.23.3, Compose Runner v0.14.5, and NeuroStore Python SDK v1.2.

2020 – Present **Lead Developer & Maintainer, NiMARE**, *Open Neuroimaging Tools Consortium*

Maintain and develop the NiMARE ecosystem for coordinate- and image-based neuroimaging meta-analysis, including sustainable releases, testing, and contributor onboarding. Recent releases include v0.22.1 and v0.22.0, with contributions spanning inference corrections, decoders, image transforms, and studyset data handling; also contribute to PyMARE v0.0.13.

Leadership Experience

2025 **Co-Organizer**, *OHBM Special Interest Group*, Brisbane, Australia
OHBM Reproducible Neuroimaging Special Interest Group

2025 – 2026 **Chair**, *OHBM Open Science Room*
Organized and moderated multiple 2026 sessions on scientific communication.

2024 **North American Technology Ambassador**, *BRIDGE*, Austin, TX
Brain Research International Data Governance Exchange

Teaching Experience

2025 **OHBM Educational Workshop Organizer**, *Brisbane*, Australia
OHBM Educational Workshop

2023 **Workshop Organizer**, *Marburg University*, Marburg, Germany
Reproducible Neuroimaging Workshop

2023 **OHBM Educational Workshop Organizer**, *Montreal*, Canada
OHBM Educational Workshop

2020 **ABCD Teaching Assistant**, *Florida International University*, Miami, FL
ABCD Educational Workshop. Organizer: Angela Laird

2019 **Workshop Instructor**, *University of Iowa*, Iowa City, IA
Brain Research Workshop. Organizer: Jan Wessel

2018 **Teaching Assistant**, *University of Iowa*, Iowa City, IA
PSY:6280:0001 Structural Functional MRI Methods/Applic. Professor: Michelle Voss

2016 **Teaching Lab Assistant**, *University of Iowa*, Iowa City, IA
ACB:6252:0001 Functional Neuroanatomy. Professor: Justin Sipla

Awards and Fellowships

2020 Neuroscience Program Gloria Lee Service Award

2019 Reproducible Neuroimaging Fellowship

2016 Post Comprehensive Exam Research Award

2015 University of Iowa Training Grant T-32

2013 Grinnellink Grant Fund

2012 Grinnell Summer Research Grant

Award Applications

2026 Application submitted for the NIH Research Specialist Award (R50)

Hackathons

2020 **Computational Psychiatry Hackathon**, *University of Iowa*, Iowa City, Iowa
Predicted bipolar diagnosis using brain measures

2019 **OHBM Hackathon**, *Rome*, Italy

2019 **Coastal Coding**, *Florida International University*, Miami, Florida
Development of meta-analysis package NiMARE

2019 **ABCD Neurocognitive Prediction Challenge**
Predicted fluid intelligence from T1w images

2018 **HACKUIOWA**, *University of Iowa*, Iowa City, Iowa
Predicted behavioral outcomes from lesions

Training

2020 **TILE (transform, interact, learn, engage) Classroom Training**, *University of Iowa*, Iowa City, Iowa

2020 **Certified Instructor**, *Software Carpentry*

2018 - 2020 **Writing in the Neurosciences**, *University of Iowa*, Iowa City, Iowa
Monthly lecture and workshop on writing samples, giving and recieving peer review

2018 – **Teaching Club**, *University of Iowa*, Iowa City, Iowa
Bi-weekly meetings about teaching strategies in the classroom

2018 **Neurohackademy**, *University of Washington*, Seattle, Washington

2017 **Code Sprint**, *Stanford University*, Stanford, California

Publications

Peer Reviewed Journals

[1] Kent J, Lee N, Salo T, Bottenhorn K, Dockes J, Blair R, Nichols T, Laird A, Poline JB, Yarkoni T, De La Vega A, Robertson G, and Ramage A (May 2026). Neurosynth Compose: A platform

for transparent and reproducible meta-analyses. *JoCN Forum*. doi:10.21428/8e6ba8ef.ad3a3f80.

[2] Kent JD, Lee N, Laird AR, Salo T, Peraza J, Bottenhorn KL, Oudyk K, Nichols TE, Poline JB, and de la Vega A (01 2026). Neurosynth Compose: A web-based platform for flexible and reproducible neuroimaging meta-analysis. *Imaging Neuroscience*, 4:IMAG.a.1114. doi:10.1162/IMAG.a.1114.

[3] Salo T, Yarkoni T, Nichols TE, Poline JB, Bilgel M, Bottenhorn KL, Eickhoff SB, Jarecka D, Kent JD, Kimbler A, Nielson DM, Oudyk KM, Peraza JA, Pérez A, Reeders PC, Yanes JA, and Laird AR (2023). NiMARE: Neuroimaging Meta-Analysis Research Environment. *Aperture Neuro*, 3. doi:10.52294/001c.87681.

[4] Esteban O, Ciric R, Finc K, Blair RW, Markiewicz CJ, Moodie CA, Kent JD, Goncalves M, DuPre E, Gomez DEP, Ye Z, Salo T, Valabregue R, Amlien IK, Liem F, Jacoby N, Stojić H, Cieslak M, Urchs S, Halchenko YO, Ghosh SS, De La Vega A, Yarkoni T, Wright J, Thompson WH, Poldrack RA, and Gorgolewski KJ (2020). Analysis of task-based functional MRI data preprocessed with fMRIPrep. *Nature Protocols*, 15(7):2186–2202. doi:10.1038/s41596-020-0327-3.

[5] Esteban O, Markiewicz CJ, Blair RW, Moodie CA, Isik AI, Erramuzpe A, Kent JD, Goncalves M, DuPre E, Snyder M, Oya H, Ghosh SS, Wright J, Durnez J, Poldrack RA, and Gorgolewski KJ (2019). fMRIPrep: a robust preprocessing pipeline for functional MRI. *Nature Methods*, 16(1):111–116. doi:10.1038/s41592-018-0235-4.

[6] Kent J and Herholz P (2019). NiBetaSeries: task related correlations in fMRI. *Journal of Open Source Software*, 4(41):1295. doi:10.21105/joss.01295.

[7] Lee HK, Kent J, Wendel C, Wolinsky F, Foster E, Merzenich M, and Voss M (2019). Home-Based, Adaptive Cognitive Training for Cognitively Normal Older adults: Initial Efficacy Trial. *The Journals of Gerontology: Series B*. doi:10.1093/geronb/gbz073.

[8] DuBose LE, Voss MW, Weng TB, Kent JD, Dubishar KM, Lane-Cordova A, Sigurdsson G, Schmid P, Barlow PB, and Pierce GL (2017). Carotid Beta-stiffness index is associated with slower processing speed but not working memory or white matter integrity in healthy middle-aged/older adults. *Journal of Applied Physiology*, 122(4):868–876. doi:10.1152/japplphysiol.00769.2016. PMID: 28126907.

Preprints

[9] Esteban O, Ciric R, Finc K, Blair R, Markiewicz CJ, Moodie CA, Kent JD, Goncalves M, DuPre E, Gomez DEP, Ye Z, Salo T, Valabregue R, Amlien IK, Liem F, Jacoby N, Stojić H, Cieslak M, Urchs S, Halchenko YO, Ghosh SS, De La Vega A, Yarkoni T, Wright J, Thompson WH, Poldrack RA, and Gorgolewski KJ (2020). Analysis of task-based functional MRI data preprocessed with fMRIPrep. *bioRxiv*. doi:10.1101/694364. Preprint version; published in Nature Protocols, 15(7), 2186–2202.

Presentations

Talks

[10] Kent J. Neurosynth Compose: From Mentions to Questions. ReproNim, October, 2026. (Webinar).

[11] Kent J, Lee N, Salo T, Bottenhorn K, Dockès J, Blair R, Nichols T, Laird A, Poline J, Yarkoni T, and de la Vega A. Neurosynth Compose: New Frontiers in Neuroimaging Meta Analyses. OHBM, Brisbane, Australia, June, 2025. (Oral).

[12] Kent J. NeuroSynth-Compose: A Web-Based Platform For Reproducible Neuroimaging Meta-Analysis. Cognition, Brain, and Behavior Seminar Series, University of Texas at Austin, Austin, TX, March, 2023. (Invited seminar).

[13] Kent J and Voss M. Measuring Communication between Brain Regions. University of Iowa Neuroscience Program, Iowa City, IA, September, 2019. (Oral).

[14] Kent J and Voss M. The Nuisance of Denoising fMRI. University of Iowa Neuroscience Program, Iowa City, IA, September, 2018. (Oral).

[15] Kent J and Voss M. Synchrony is Key for Aging and Cognitive Control. University of Iowa Neuroscience Program, Iowa City, IA, October, 2017. (Oral).

[16] Kent J and Voss M. Midbrain Activity during Task and Rest: Two Sides of the Same Coin? University of Iowa Neuroscience Program, Iowa City, IA, May, 2016. (Oral).

[17] Kent J and Queathem E. Affective and Psychological effects of simulated outdoor exercise. Midwestern Psychological Association, Chicago, IL, May, 2015. (Oral).

[18] Kent J and Vaidya J. Reward Sensitivty and Attentional Control in Adolescents. University of Iowa Neuroscience Program, Iowa City, IA, May, 2015. (Oral).

[19] Kent J and Voss M. Brain Stripping Algorithms, Betcha can't use just one. University of Iowa Neuroscience Program, Iowa City, IA, June, 2015. (Oral).

Poster Presentations

[20] Kent J, Lee N, Salo T, Bottenhorn K, Dockès J, Blair R, Nichols T, Laird A, Poline J, Yarkoni T, and de la Vega A. Neurosynth Compose: New Frontiers in Neuroimaging Meta Analyses. Human Brain Mapping, Brisbane, Australia, June, 2025. (Poster).

[21] Kent J, Lee N, Salo T, Bottenhorn K, Dockès J, Blair R, Nichols T, Laird A, Poline J, Yarkoni T, and de la Vega A. NeuroSynth Compose: A Web-Based Platform for Reproducible Meta-Analyses. Human Brain Mapping, Seoul, South Korea, June, 2024. (Poster).

[22] Kent J, Lee N, Salo T, Bottenhorn K, Dockès J, Blair R, Nichols T, Laird A, Poline J, Yarkoni T, and de la Vega A. Neurosynth Compose: A Web-Based Platform to Reproducibly Create, Replicate, and Extend Meta-Analyses. Human Brain Mapping, Montreal, Canada, July, 2023. (Poster).

[23] Kent J, Lee N, Salo T, Bottenhorn K, Dockès J, Blair R, Nichols T, Laird A, Poline J, Yarkoni T, and de la Vega A. Neurosynth-Compose: A Web-Based Platform For Reproducible Neuroimaging Meta-Analysis. Human Brain Mapping, Glasgow, Scotland, June, 2022. (Poster).

[24] Kent J, Evans J, Magnotta V, Vaidya J, Shaffer J, and Koscik T. Big Brain Data: New Opportunities at the University of Iowa. University of Iowa Informatic Program, Iowa City, IA, February, 2019. (Poster).

[25] Kent J, Lee HK, Wendel C, Wolinksy F, Foster E, Merzenich M, and Voss M. Task Related Brain Connectivity Decreases After Cognitive Training. Cognitive Neuroscience Society, San Francisco, CA, March, 2019. (Poster).

[26] Kent J and Voss M. NiBetaSeries: Tasking State Correlations. Human Brain Mapping, Rome, Italy, June, 2019. (Poster).

[27] Kent J, Lee HK, Schultz E, Foster E, Wolinksy F, Merzenich M, and Voss M. Can adaptive cognitive training improve efficiency of attentional control in the aging brain? Society for Neuroscience, San Diego, CA, November, 2016. (Poster).

Software

Contributions to Open Source Scientific Software

[28] Kent JD, Rokem A, VanderPlas J, Holdgraf C, DeStasio K, Ludwig R, Martin R, Toro-Serey C, Eschenburg K, Erramuzpe A, Herholz P, Blue B, Whitaker K, and Voss M (January 2019). HBClab/NiBetaSeries: v0.2.3. doi:10.5281/zenodo.2552303.

[29] Esteban O, Blair R, Markiewicz CJ, Berleant SL, Moodie C, Ma F, Isik AI, Erramuzpe A, Kent JD, Goncalves M, DuPre E, Poldrack RA, and Gorgolewski KJ (January 2018). poldracklab/fmriprep: 1.0.5. doi:10.5281/zenodo.1156273.

[30] Gorgolewski KJ, Esteban O, Ziegler E, Notter MP, Burns C, Ellis DG, Johnson H, Manhães-Savio A, Hamalainen C, Jarecka D, Yvernault B, Markiewicz CJ, Salo T, Waskom M, Clark D, Wong J, Loney F, Madison C, Modat M, Goncalves M, Dewey BE, Clark MG, Dayan M, Keshavan A, Clark D, Pinsard B, Gramfort A, Berleant S, Nielson DM, Varoquaux G, Cipollini B, Nielson DM, Markello R, Rokem A, Wassermann D, Halchenko YO, Visconti di Oleggio Castello M, Kaczmarzyk J, Moloney B, Hanke M, Horea C, Bougacha S, de Hollander G, Mordom D, Buchanan C, Tungaraza R, Mancini M, Pauli WM, Iqbal S, Sikka S, Schwartz Y, Malone IB, DuPre E, Frohlich C, Welch D, Forbes J, Watanabe A, Cumba C, Huntenburg JM, Eshaghi A, Kastman E, Ginsburg D, Schaefer A, Acland B, Kent J, Perez-Guevara M, Gillman A, Bougacha S, Forbes J, DuPre E, Nichols BN, Giavasis S, Jordan K, Kleesiek J, Erickson D, Küttner R, Haselgrove C, Correa C, Ghayoor A, Millman J, Haehn D, Lampe L, Lai J, Zhou D, Blair R, Glatard T, Renfro M, Liu S, Kahn AE, Triplett W, Jordan K, Stadler J, Kong XZ, Hallquist M, McDermottroe C, Chetverikov A, Salvatore J, Mertz F, Park A, Poldrack R, Jordan K, Craddock RC, Inati S, Hinds O, Cooper G, Perkins LN, Tambini A, Marina A, Mattfeld A, Noel M, Snoek L, Matsubara K, Cheung B, Gerhard S, Rothmei S, Urchs S, Pannetier N, Durnez J, Geisler D, Floren A, Sharp P, Mancini M, Weinstein A, Broderick W, Saase V, Andberg SK, Harms R, Khanuja R, Schlamp K, Arias J, Visconti di Oleggio Castello M, Papadopoulos Orfanos D, Tarbert C, De La Vega A, Nickson T, Brett M, Dubois M, Falkiewicz M, Podranski K, Pellman J, Linkersdörfer J, Varada J, Schwabacher I, Flandin G, Liem F, Pérez-García F, Shachnev D, McNamee D, Davison A, and Ghosh S (January 2018). nipy/nipype: Nipype - v1.0.0. doi:10.5281/zenodo.1160610.

Selected Media and Interviews

How AI Maps Brain Activity from Text Across 30,000 Studies. Faculti. Discussion of NeuroVLM and Neurosynth Compose.

From Data to Discovery: James Kent Pioneers Next-Generation Neuroimaging Meta-Analyses. University of Iowa Neuroscience Graduate Program, 2026.

Digital Outreach

2025 – Present **Channel Administrator and Contributor**, *NeuroSynth Organization*
Administer and publish project updates through the organization's YouTube, Bluesky, and LinkedIn channels.

Service/Leadership

2018 – 2020 **Organizer**, *Hacky Hour*, University of Iowa, Iowa City, IA
I work with individuals to solve their programming/data problems/encourage community across departments

2018 – 2020 **Organizer**, *Neuroimaging Workgroup*, University of Iowa, Iowa City, IA
Curate, schedule, and sometimes lead monthly sessions about the latest tools/methods to analyze MRI data

2018 – 2020 **Organizer**, *BrainHack@UI*, University of Iowa, Iowa City, IA
Secured funding, space, materials, and a committee to create a yearly brainhack

2015 – 2020 **Committee Member**, *Eastern Iowa Brain Bee*, University of Iowa, Iowa City, IA
Presented at local high schools to recruit participants and helped with various aspects of setup and execution of the Brain Bee

2015 **Student Member**, *YOU@UI Neuroscience Outreach*, University of Iowa, Iowa City, IA

2015 **Demonstrator**, *Brain Awareness Week Outreach*, University of Iowa, Iowa City, IA

Contributor, *BIDS (Brain Imaging Data Structure)*

Technical Expertise

Research software engineering for reproducible neuroimaging and meta-analysis: platforms, libraries, and data standards; translating research workflows into tested, maintained community tools.

Technologies

Scientific	FreeSurfer, FSL, AFNI, NiPype, scikit-learn, PsychoPy, NiMARE
Software/Web Development	Flask, PostgreSQL, NGINX, Docker, Continuous Integration (GitHub Actions, Travis CI), OpenAPI/RESTful APIs,
Cluster computing	Oracle/Sun Grid Engine (SGE)
Version control	git (GitHub, git-annex)

Programming Languages

Proficient	Python 2/3, Shell
Competent	R, LaTeX
Familiar	MATLAB, JavaScript

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